LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 7: Randomness and Probability

Module 3 | 45 minutes | Random simulation | Grades 6–8 · ages 11–14

Objective: Use random numbers to simulate chance events many times, and compare experimental results with theoretical probability.

Success criteria

- I can simulate a random event many times.
- I can compare results with theory.
- I can explain the law of large numbers.

Key vocabulary

- **simulation**: A program that imitates a real process.
- **trial**: One run of an experiment.
- **probability**: How likely something is (0 to 1).
- **law of large numbers**: More trials → results closer to theory.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Coin relay (5 min): each pair flips 10 coins; combine class totals on the board.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Coin-flip lab: learners follow the steps.
14 min	**Perform**	Code Lab: flip 100 coins, Small samples
9 min	**Probe**	Code Lab: random walk, activity
4 min	**Wrap-up**	Exit ticket: Name one thing that happens as you increase the number of trials.

Unplugged warm-up

Coin relay (5 min): each pair flips 10 coins; combine class totals on the board.

Teaching the idea

Monte Carlo-style simulation: repeated random trials approximate probabilities. Experimental frequencies converge to theoretical probability as trials increase (law of large numbers). Pseudo-randomness is sufficient for simulation.

Analogy: Polling: ask enough people and the result reflects the whole population.

Live demo: Run 10 flips several times (wild results), then 1000 flips (stable).

Questions to ask

Ask	Listen for
Why isn't it exactly 50?	Randomness varies.
What happens with more flips?	Closer to 50%.
What else could we simulate?	Dice, weather, queues.

Differentiation

Support
Provide the loop; learners write the condition.

Stretch
Simulate two dice 1000 times and count how often the total is 7.

Common misconceptions

Misconception	How to fix it
After 5 tails, heads is due	Each flip is independent (gambler's fallacy).
Computers can't do randomness	Pseudo-random is fine for simulation.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Name one thing that happens as you increase the number of trials.

Homework
Flip a real coin 20 times and compare with your simulation.

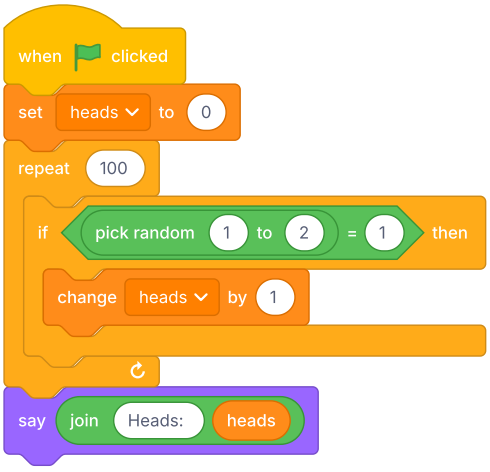
Answer key — Kitabash activities

1. Code Lab: flip 100 coins (perform · code lab)

Goals checked: Flips randomly · More than 30 heads · Fewer than 70 heads · Reports it

One solution:

Scientist



2. Small samples (perform · multiple choice)

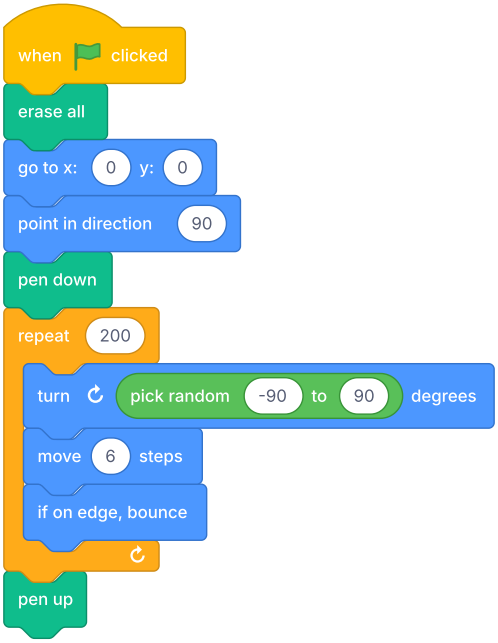
Answer: That can happen by chance; try many more flips

3. Code Lab: random walk (probe · code lab)

Goals checked: Random turns · A long path · Stays on stage

One solution:

Bug



4. Explain your thinking (probe · free response)

Look for: random, flips, closer. **Example:** I got 47, 55, 50, 44 and 52 heads. They are different because each flip is random. With 10,000 flips the percentage of heads would be much closer to 50% because the law of large numbers evens out the luck.

Answer key — worksheet

1. **Key words:** simulation = A program that imitates a real process. · trial = One run of an experiment. · probability = How likely something is (0 to 1). · law of large numbers = More trials → results closer to theory.
2. **Plan your code:** Goals: Flips randomly; More than 30 heads; Fewer than 70 heads; Reports it.
3. **Small samples:** A. That can happen by chance; try many more flips — Small samples vary a lot. The law of large numbers needs many trials.
4. **Plan your code:** Goals: Random turns; A long path; Stays on stage.
5. **Explain your thinking:** I got 47, 55, 50, 44 and 52 heads. They are different because each flip is random. With 10,000 flips the percentage of heads would be much closer to 50% because the law of large numbers evens out the luck.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.